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Manifestations of Combustion Instability in a Multi-swirl Stabilized Gas Turbine Combustor

Guoqiang Li* and Ephraim J. Gutmark†
University of Cincinnati, Cincinnati, OH, 45220

This paper is on study of manifestations of combustion instability in a multiple swirl low emission gas turbine combustor. Unstable combustion was investigated in different combustor setup, including with or without premixing section, and with gaseous and liquid fuel via measurement of dynamics of pressure, flame, and radical emissions. A set of four optical fibers integrated inside the multiple swirler fuel injector assembly monitored the local active radical emissions to provide phase information on unsteady flame dynamics. The acoustic pressure signal from a microphone is conditioned as the trigger signal for phase-locked OH* chemiluminescene imaging of the global flame. The flame dynamics are shown to be in different format for different geometries and types of fuel. The motions of flame front are shown to be closely related to the combustion instability driving mechanism, either periodic large flow structures or pulsation in fuel line. Analysis on transition from unstable to stable combustion reveals that the advantage of better mixing with gaseous fuel is offset by the faster growth rate of pressure oscillation. It was also found in this paper that the oscillations of pressure lead flame in both gaseous fuel and spray combustions when unstable combustion takes place.

Nomenclature

D	=	Diameter of the mixer exit
L_{mt}	=	Nondimensional length of mixing tube
f	=	frequency
m_a	=	mass flow rate of inlet air
p_3	=	pressure of inlet chamber (gauge pressure)
p'	=	oscillation of acoustic pressure
T_3	=	temperature of inlet air
α	=	angle of outer swirler
β	=	angle of intermediate swirler
γ	=	angle of inner swirler
Φ	=	fuel equivalence ratio

I. Introduction

Combustion instability, which is recognized as the coupling of unsteady heat release and acoustic oscillations and often manifested as high magnitude acoustic noise, challenges the operations of conventional diffusion combustion systems and, more severely, modern lean burn combustion systems. The driven mechanism of combustion instability, or more exactly, thermo-acoustic combustion instability, can be categorized into two groups according to Mongia et al[1]. First category, which is often associated with high frequency (>1 kHz), is the coupling between the pressure oscillation and the instantaneous flame position and shape as the flame surface responds to the pressure disturbance. The second is the coupling between oscillations of pressure and local fuel/air ratio, with

* Research Assistant Professor, Department of Aerospace Engineering, University of Cincinnati, AIAA member
† Ohio Eminent Scholar, Department of Aerospace Engineering, University of Cincinnati, AIAA associate fellow.

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frequency ranged from 100 to 1000 Hz. Several sub-mechanisms can affect local fuel/air ratio, including fuel line/air feeding perturbation and unsteady flow structures. Also, according to the source terms, i.e. which one comes first as to heat flux and pressure oscillation, Ducruix et al.[2] classify the driven mechanisms into: flame/boundary interaction, flame/vortex interaction, in which heat flux leads pressure oscillation, flame response to acoustic wave, and unsteady strain rate and equivalence ratio fluctuation, in which these unsteady parameters cause pressure or/and velocity fluctuation and finally heat release fluctuation. These reviews expose the complexity of combustion instability driving mechanisms and the possibility of interaction between different mechanisms. Therefore, it is hard to isolate completely one driven mechanism from the other among all these mechanisms. For instance, the oscillation in fuel/air equivalence is possibly caused by dynamics of large vortical structures and at the same time associated with flame/vortex interaction[2], and consequently results in oscillations of heat release.

Identification of combustion instability mechanisms requires detailed experimental measurements of oscillations in flame, flow structure, and pressure, combined with modeling effort. The manifestation of combustion instability in flame and acoustic fields may differ in great degree depending on the geometric characteristics of combustion system. Chemiluminescence imaging of active radicals, such as CH*, OH* and C^2, are extensively used in characterizing dynamics of flame in unstable combustion of various burners[3-5]. For a dump combustor, Poinsot et al.[3] measured the dynamics of the vortices shed from the dump plane and periodic flame evolution with C_2 chemiluminescene by phase-locking with the pressure oscillation signal, and concluded that large vortices shed from the sump plane caused flame periodic change when traveling downstream and were responsible for the combustion instability. Gas turbine combustion systems extensively employ swirlers to stabilize the flame and enhance the fuel/air mixing, so called swirl stabilized combustor, and form a more complicated flow and combustion fields inside the combustor domes as compared with that of dump combustors. Paschereit et al.[4] conducted investigattion and control of unstable thermo-acoustic instability in an experimental low-emission swirl stabilized combustor, and observed several axisymmetric and helical unstable modes for fully diffusion and premixed combustion by measurement of phase relationship between radical signals detected by two optical fibers on the radial direction and phase-locked OH* imaging. These unstable modes were identified to be associated with flow instabilities related to the recirculation wake-like region on the combustor axis and shear layer instabilities at the sudden expansion. These investigations highlight the application of optical diagnostic technology in resolving unstable combustion flame and heat release. Besides radical chemiluminescence, application of Planar Laser Induced Florence (PLIF) [6] and high speed direct imaging of the flame luminance [7] in detection of unstable flame have also been reported. Comparison of PLIF and OH* emission imaging for unstable combustion [6] suggested that CH LIF and OH* emission were suitable indicator for the average shape and locations of flame front.

Despite numerous investigations on combustion instability, less conclusive comments can be firmly made on the exact driving process of combustion instability because of the complicated dynamics of flame, pressure, flow structure, fuel spray, and mixing and possible disturbances form fuel, air, equivalence ratio that may be somehow involved in unstable combustion. This study, based on a design of multiple swirl low emission gas turbine combustor, employed extensively experimental instrumentation to simultaneously monitor the acoustic pressure, local and global radical emissions under unstable combustion of different geometry configurations and types of fuels. Data presented in this paper will provide useful information on coupling of flame and pressure, motion of flame front, and transition from unstable to stable combustion.

II. Experimental Setup

The experimental tests were performed in an atmospheric combustion test rig that simulates a multiple swirler Lean Direct Fuel Injection (LDI) gas turbine combustor (Fig.1a). The rig is set up vertically with the combustion air preheated by a 36 kW electrical heater to the desired temperature entering through the inlet flange. The test rig consists of an inlet flange, flow conditioning chamber, plenum chamber, and the combustion chamber. The flow conditioning chamber and plenum chamber is constructed from 6 inches schedule 80 stainless steel pipes with inner diameter 133.4 mm (5.25") and total length 978mm (38.5"). In the flow conditioning chamber, the flow passes through a perforated cone and then through a sequence of 5 fine-mesh screens whose mesh size is gradually reduced to 0.05mm (0.002") diameter with 44% open area ratio. The multiple swirler and fuel injector assembly is centered on a mounting plate (6.35mm=0.25" thick) and flush mounted with the combustor chamber inlet. The combustion chamber is a 457 mm (18") long cylindrical quartz tube, with 101.6 mm (4") inner diameter, and 2.5 mm wall thickness. The flame inside this tube is fully observable by an Intensified CCD camera for flame emission imaging.

The core of the LDI combustor is the Triple Annular Research Swirler (TARS), which was developed by Delavan Gas Turbine Products (DGTP, a division of Goodrich Aerospace) in collaboration with General Electric

Aerospace Engines (GEAE). TARS (Fig. 1b) features three air passages, each with an individual swirler. Different configurations of TARS can be designed by changing the swirlers---outer swirler α, intermediate swirler β, and inner swirler γ--- either to different swirling angles or different rotating directions, to form different swirling flow fields. The inner and intermediate swirlers are axial swirlers whereas the outer swirler is radial. Hereafter, the swirler configuration will be labeled with the swirler angle in the order of outer, intermedicate and inner, and with "C" labeling the counter-rotating swirl direction. For example, S5545C45 means that the swirler configuration has 55º outer swirler, 45º intermediate swirler in counter-rotating direction and 45º inner swirler. The total length of TARS is 66 mm (2.6") and the diameter of the outer swirler exit is 50.8 mm (2"). The intermediate swirler is located 66 mm (2.6") upstream of TARS exit whereas the inner swirler is at 50 mm (2 ") upstream of TARS exit plane. S5545C45 was the swirler configuration for all the data presented in this paper. Two sets of distributed fuel injection circuits are integrated within the TARS assembly: pilot fuel circuit injects fuel inward into the intermediate swirling air passage and the main fuel circuit injects fuel outward into the outer swirling air passage. These two fuel circuits are separated, and controlled and metered individually. The number of injection holes for the pilot and main fuel is four and eight, respectively. All the injection holes are identical, 0.762mm (0.03") in diameter. These injection holes are perpendicular to the wall so that the fuel stream is injected normal to the air streams.

A short mixing tube (Fig. 1c) can be inserted between the TARS exit and the combustor dump plane. This mixing tube has the same inner diameter as the TARS exit. The total length of the mixing section, which is composed of the mixing tube, sealing gasket, and the mounting plate, is normalized with the diameter of TARS exit, D, as L_{mt}.

A set of four multiple mode optical fibers were integrated inside TARS for monitoring the dynamics of flame (Fig. 1d). Three fibers are located circumferentially on the expansion cone of the outer swirler on the 45mm diameter circle with interval angle of about 120º. The fourth one is installed at the center the inner swirler, aligned with the centerline of the TARS and the combustion chamber. The fibers are label as 1, 2, 3, and 4, respectively, as shown in Fig.1d. All the fibers were pointing vertically up at the flame with about 1mm recession from the cone face. The light emitted by the flame was received by the optical fibers in a narrow cone angle. The light is filtered by narrow band pass filters, then amplified and converted to voltage signal by a bundle of photomultiplier tubes (PMT).

A Brüel & Kjær microphone (Model 4939-A-011) was set up 0.5m away from the combustor to detect the acoustic emissions. Signal from this sensor was used by Digital Signal Processing (DSP) board to generate a phase-locked trigger signal for OH* chemiluminescence imaging. The flame radical imaging is performed by an intensified CCD camera (Roper Scientific Super Blue 1024x1024 pixels, 12-bit) with timing generator that can handle minimum about 2ns time delay. This camera was set up in sequential gate mode that was phase-locked with the pressure dynamic signal for the microphone. Narrow band-pass filters, 310.02 nm center wavelength with FWHM 10 nm (Andor 310FS10-50), was used for OH* imaging to detect the strong OH*-line ($A^2\Sigma+$, $v'=0 \rightarrow X^2\Pi$, $v''=0$, $\lambda=306.4$ nm).

Static pressure transducer (Druck PMP 4000 series, ±0.04% FS accuracy, 1 psi range) and type K thermocouple were mounted on the wall of the combustor inlet section to monitor the pressure drop across TARS dynamics pressure oscillations in the inlet section, and inlet air temperature, respectively. The air mass flow rate was meterd by a digital flow meter (Eldridge Prouducts, Inc, 1% accurancy). The liquid fuel flow rates of pilot and main fuel were metered by two turbine flow meters and the gas fuel flow rate by a laminar type digital gas flow meter. To evaluate the influence of unstable combustion on emissions, an extractive gas sampling and analyzing system (from California Analytical Instrument) was utilized to continuously monitoring oxides of nitrogen (NO_x), carbon monoxide (CO), oxygen (O_2), and carbon dioxides (CO_2). The system has 10 ppb resolution for NO/NO_x, 0.1 ppm for CO, 0.1% for CO_2 and O_2. The sampling probe is heated to above 100ºC to eliminate water vapor from the sample. The CO analyzer is designed to be in a serial arrangement using a pneumatic type differential sensor to minimize the interference from CO_2 or water vapor.

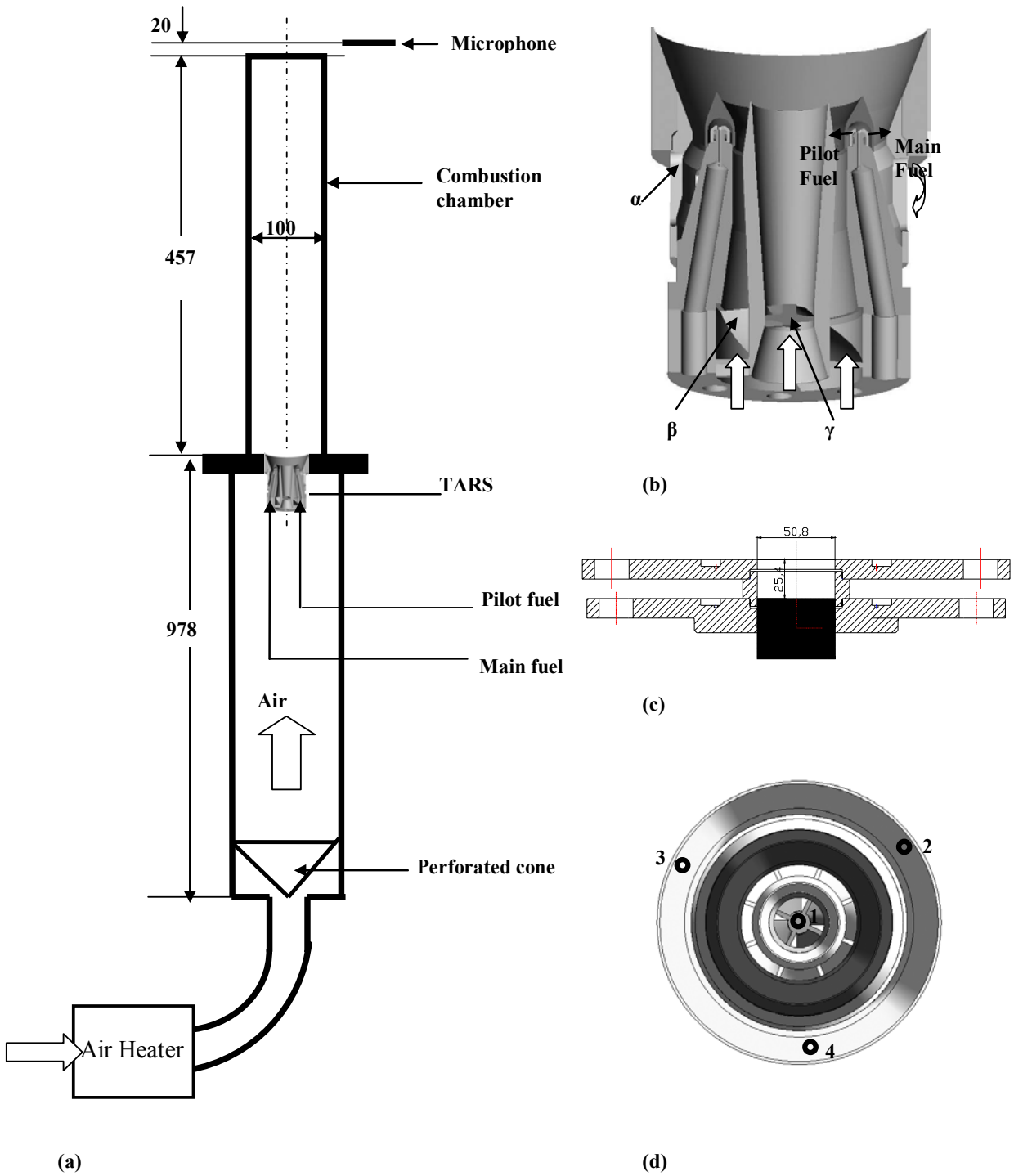


Figure 1: Schematic drawing of (a) atmospheric test rig, (b)Triple Annular Research Swirler (TARS), (c) mixing section, and (d)location of OH optical fibers . The dimensions are in mm.

All process parameters, including main air flow rate (m_a), air inlet temperature (T_3,), plenum pressure (p_3), pressure inside combustion chamber (p_4), combustion gas temperature (T_4), combustor wall temperature (T_w), total fuel flow rate (W_f), were recorded by Labview software, as well as emission data from the gas analyzer. The dynamic signals of acoustic and light emissions are recorded at 5 kHz for each channel. Four cases are tested in this study (Table 1) for investigating the manifestations of combustion instability with respect to acoustic pressure, flame, the intensity of local active radicals at different locations, and the influence of combustion instability on CO and NO_x emissions. The first two cases are with gaseous propane: one has no mixing tube (L_{mt}=0) and the other has

25.4mm long premixing section (L_{mt}=0.5). The third and fourth cases are with liquid fuel at 50/50 fuel distribution through the pilot and main fuel circuits. The difference between case 3 and 4 is that case 3 used Jet-A fuel and case 4 Ethanol. In all these cases, the inlet air was preheated to T_3=230ºC.

Table 1: Experimental setup and operating conditions

Cases	P_3 (pa)	T_3 (ºC)	m_a (kg/s)	Φ	L_{mt}	Fuel	f (Hz)
1	0.54	230	0.023	1.0	0	Propane	386
2	0.36	230	0.023	0.75	0.5	Propane	357
3	0.99	230	0.032	0.6	0	Jet-A	357
4	0.78	230	0.032	0.68	0	Ethanol	377

III. Results and Discussion

For the multi-swirl LDI combustion system we study here, manifestation of combustion instability differ in many aspects depending on the fuel types and the insertion of mixing tube. These aspects include dynamics of flame, distribution of local radical, the initialization of combustion instability, and the relationship between combustion instability and CO and NO_x emissions. In this section, we will first present data on flame and acoustic pressure for different cases and then discuss the differences of flame dynamics and the growth rate of pressure oscillations between them.

A. Case1: Propane without mixing tube (L_{mt}=0)

For propane without mixing tube, the combustion bursts into unstable at Φ greater than 0.94. The spectra of the acoustic pressure p' is shown in Fig. 2a, with dominant frequency f at 386Hz and existence of the second and third harmonics. This fundamental frequency is very close to the quarter-wave mode frequency of 395Hz, which is calculated from the combustion chamber length and measured bulk combustion gas temperature at the exhaust, 1400K. A 4th order Butterworth filter centered at the instability frequency with 100Hz band width is used to filter raw dynamic signals of pressure and OH* from four channels of optical fibers. Comparison of waveforms of p' and OH* at the four fiber locations (Fig. 2b) shows that oscillations of OH* at the three circumferentially distributed fibers 1, 2, and 3 were almost in phase with p', whereas the OH* at the central flame region, that was denoted as OH1, led p' about one sixth circle. Fig. 3 shows the FFT spectra of each individual OH* signal. In addition to the fundamental (f=387Hz) and the secondary harmonic frequency (f=774Hz), the third harmonic (f=1161Hz) was shown in the signals of the three circumferential fibers, OH*2, 3, 4, except the OH*1 that was monitoring the OH* radicals in the central flame region. That the high frequency was not observed by the central optical fiber may indicate existence of a circumferential combustion instability that is possibly responsible for the phase difference of between OH1 and OH2, 3, and 4 in waveforms of Fig. 2b. Magnitude of this high frequency was only one third of that of the dominant frequency at 387Hz, therefore the primary oscillations of flame is still in the longitudinal direction as was clearly depicted by phase-locked OH* imaging in the next figure.

Fig. 4a shows a sequence of phase averaged OH* images at different phases within one period of the unstable combustion of Case 1. The flame starts with a quasi-flat flame front at 0° and gradually moves in the axial direction up to

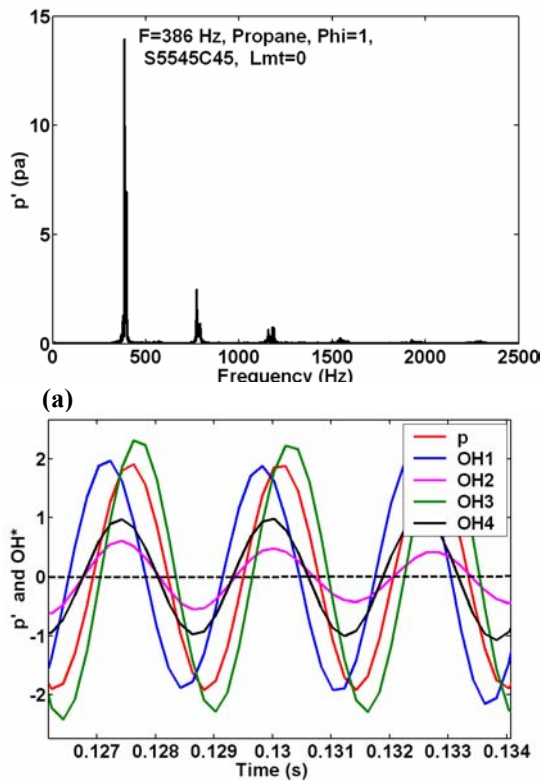


Figure 2: Combustion instability at Case 1: propane, L_{mt}=0. (a) FFT spectra of acoustic pressure p'; (b) Comparison of waveforms of acoustic pressure p' and OH* at four locations 1, 2, 3 and 4 (The waveforms shown here are filtered by band pass filters centered at 386Hz and the magnitudes of the signals are adjusted for purpose of comparison).

111°. In this stage, the flame also curves at the bottom surface as the consequence of vortex structures forming along the inner edge of the swirling jet[8]. The images at phase 92.5 ° and 111° clearly reveal this flame structure. The large vortical structures brought large amount of fuel/air mixture to the central flow region. After phase angle 111°, the intense flame region quickly expands to almost the whole flame zone from 129.5 to 166.5 ° when the mixture accumulated from the first stage gets burning. The flame starts to split at 222° and less intense flame regions were formed on the both sides of the jet. Flame length gets shorter in a slow pace from 240.5 to 351.5° when the two lopes of flame split away. Fig. 4b shows the integration of pixel intensity in each single image versus phase, which clearly depicts the periodic oscillation of the flame at 387Hz.

It was noticed that the flame front gradually moved up and down along with intensity change in the longitudinal direction in Fig. 4a, as is contrast to variation of only flame intensity observed by some researchers [5, 7]. To further emphasize this point, we tracked the location of the flame front and the flame front intensity (Fig. 5) and found that the flame front oscillated in phase with the variation of flame intensity. This is one very important indication that unstable combustion in this gaseous fuel LDI setup is related to pulsation of flame front.

Although it is commonly recognized that the coupling between unsteady heat release and pressure oscillations sustains combustion instability, few knowledge is available on the initialization phase of the unstable combustion. Through high speed imaging[7], it was observed

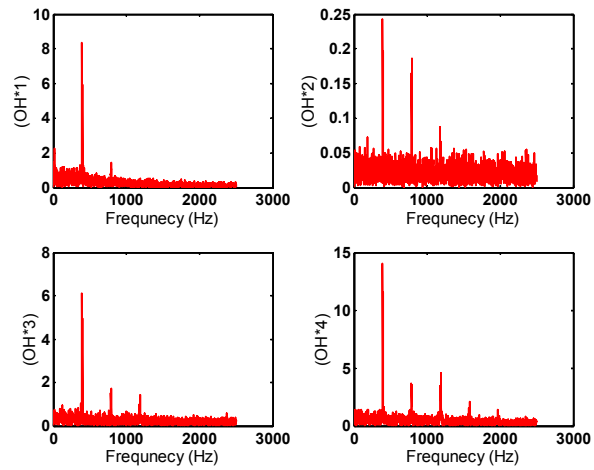
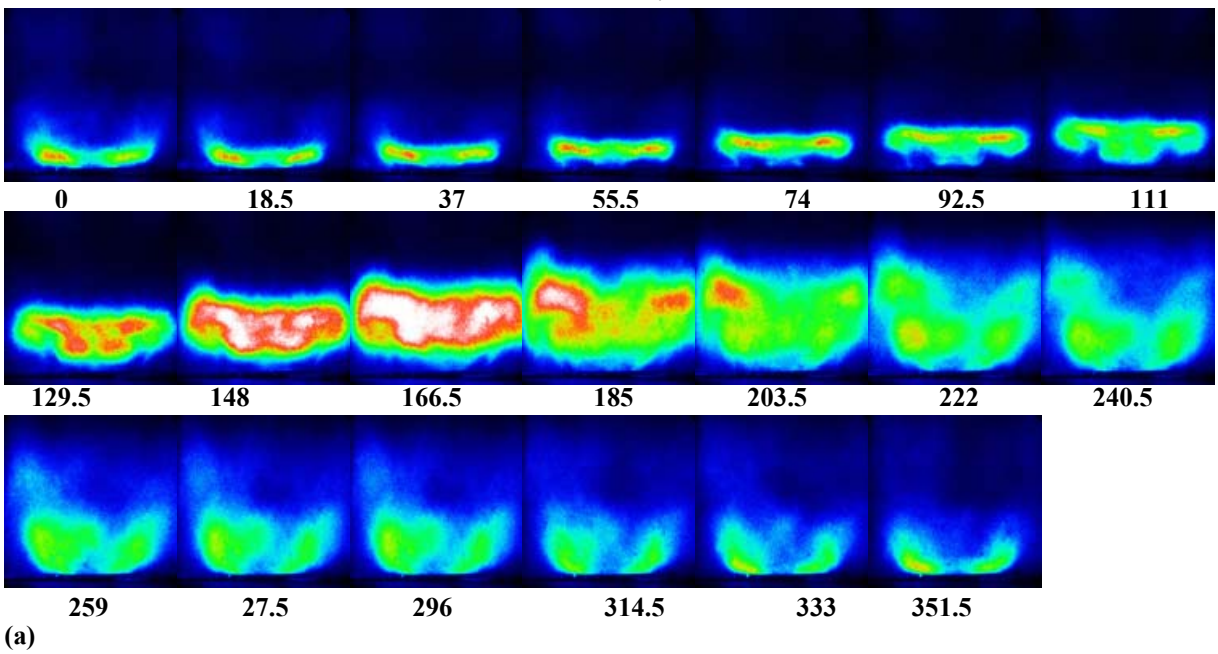
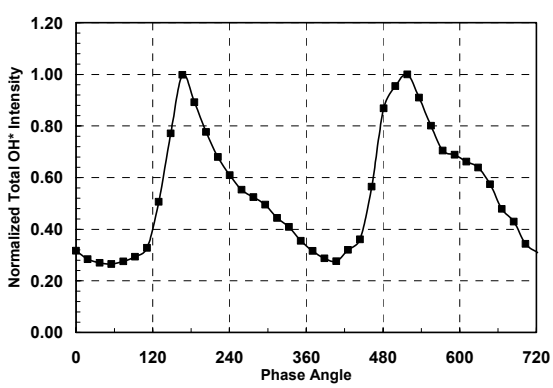


Figure 4: Spectra of OH* signals detected by optical fibers at four locations: 1, 2, 3 and 4. Case1, propane, L_{mt}=0.



(a)



(b)

Figure 3: Oscillations of flame at unstable combustion of propane, L_{mt}=0, case 1: (a) Phase averaged OH* images at different phase angles within one cycle of unstable combustion. The unit of angle is degree. (b) Normalized total OH* intensity integrated from individual image versus phase angle.

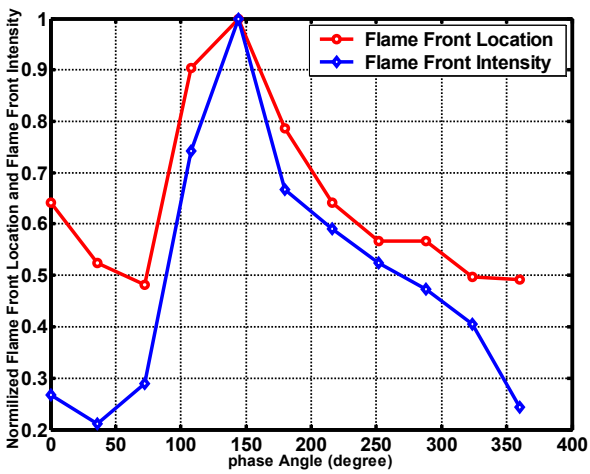


Figure 6. The dependence of flame front location and flame front OH* intensity on phase angle of propane, L_{mt}=0, Case1. The y variables are normalized with the maximum values.

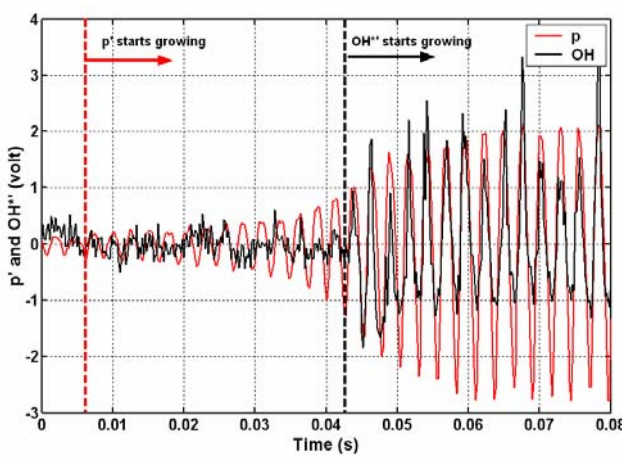
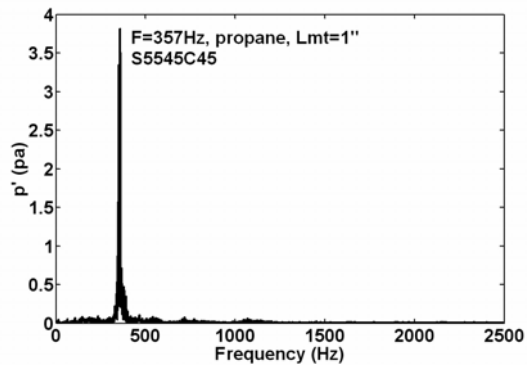
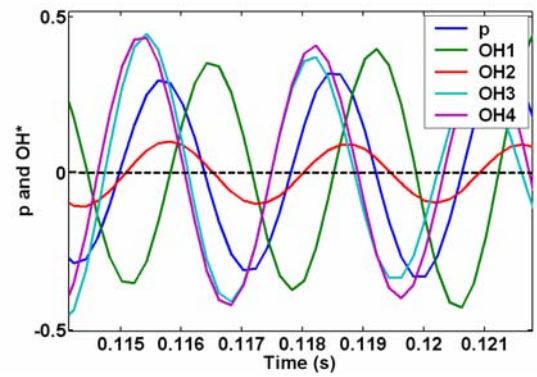


Figure 7. Time trace of p' and OH* at the initialization phase of unstable combustion: propane, L_{mt}=0, Case 1.



(a)



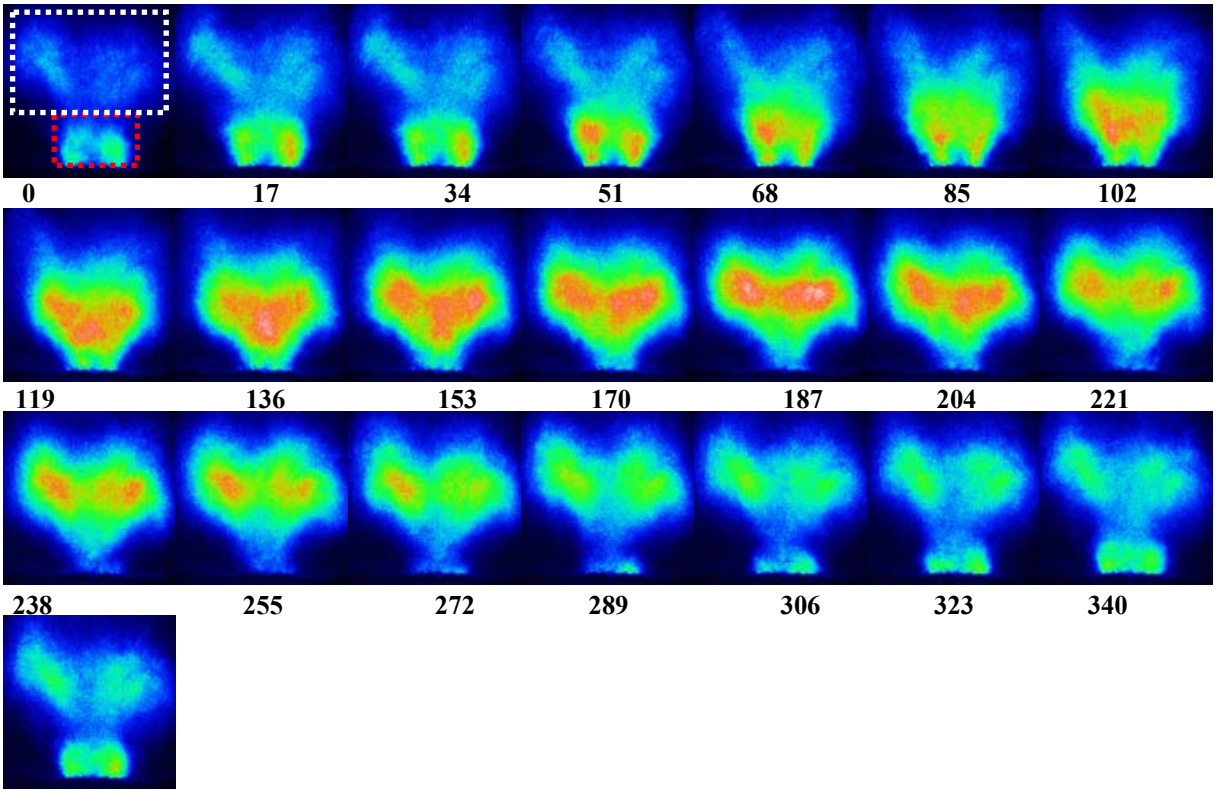
(b)

Figure 5: Combustion instability of propane, L_{mt}=0.5, Case 2: propane, L_{mt}=0.5. (a) FFT spectra of acoustic pressure p'; (b) Comparison of waveforms of acoustic pressure p' and OH* at four locations 1, 2, 3 and 4 (The waveforms shown here are filtered by band pass filters centered at 357Hz and the magnitudes of the signals are adjusted for purpose of comparison).

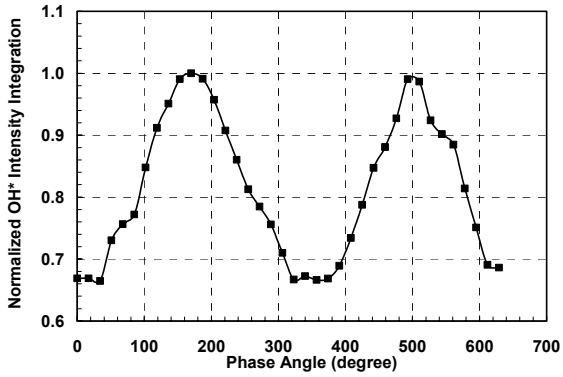
that, during the transition from stable to unstable combustion of compressed natural gas, the flame first gradually increased luminosity, then sharply burst into intense unstable flame, and finally oscillated at slightly lower intensity. Fig. 6 depicts the time trace of p' and OH* during the transition stage of unstable combustion at Case 1. The unstable combustion first initialized from growth up in oscillations of pressure (marked by the red line in Fig.6). After more than ten oscillation cycles of pressure (about 40ms), the flame suddenly burst into strong oscillation (marked as black line in Fig. 6) to complete the feedback loop of hear release and pressure oscillations. The leading oscillation of pressure indicates that for this case the flame dynamics and so the combustion instability can only be sustained when pressure oscillation grows up to certain level. It was proposed that the oscillations of pressure was associated with the perturbation from flow dynamics as is evidenced by phase-locked Particle Image Velocimetry (PIV) measurement of velocity flow field [8].

B. Case 2: Propane with mixing tube (L_{mt}=0.5)

Mixing tube promotes the occurrence of combustion instability. For same operating condition as L_{mt}=0, large magnitude instability takes place for L_{mt}=0.5 at Φ =0.73 and greater. The dominant instability frequency is 357Hz with no significant harmonics (Fig. 7a). The fundamental frequency is lower than that of L_{mt}=0 because the 1" length mixing tube increases the combustion chamber length. In the comparison of waveforms of p' and OH* of 4 channels (Fig. 7b), all other signals were approximately in phase except the signal of OH*1, which was almost out of phase with the other three optical signals.



(a)



(b)

Figure 8: Oscillations of flame at unstable combustion of propane, L_{mt}=0.5, Case 2: (a) Phase averaged OH* images at different phase angles within one cycle of unstable combustion. The unit of angle is degree. (b) Normalized total OH* intensity integrated from individual image versus phase angle.

The sequence of OH* images (Fig. 8a) shows very different behavior for unstable combustion of L_{mt}=0.5 as compared with that of L_{mt}=0. Regarding to its structure, the flame can be divided into two parts, root and stem, marked with red and white block in Fig. 8a. The root part is close to the mixing tube outlet and the stem is on top of the root with larger flame area. During the first half cycle from 0 to 187°, the flame grows up from the root, becomes more and more intense and grows into only stem part left. Then the flame weakens in the stem part until 255°. Starting from this phase, the root part becomes high intense flame along with the flame weakening in the stem part. Figure 8b shows the total OH* intensity of image versus phase angle. It clearly depicts the flame oscillation periodically repeats with pressure. The difference in phase angle of OH* signals (Fig. 7b) can be explained by the oscillation of flame in root and stem parts. The flame intensity from root and stem parts is integrated separately and plotted in Fig. 9. OH*1 detected the flame in Central Recirculation Zone (CRZ), which mainly consisted the root flame, and captured the strongest oscillation at the about 357º. All the other three circumferential fibers detect the flame of stem part that is most intense at 187º. The OH* from the root and the stem is shown clearly out of phase.

Because the mixing tube significantly enhanced the axial component of swirling flow[10] and made the combustor geometry more similar to dump combustor, the oscillation in flame of L_{mt}=0.5 is possibly caused by periodic change in jet velocity, which has been proposed by Poinsot et al.[3] to explain vortex driven combustion instability for dump-combustors. The single dominant frequency indicates more concentrated energy distribution that may be associated with the promotion of large coherent structures by the mixing tube[10].

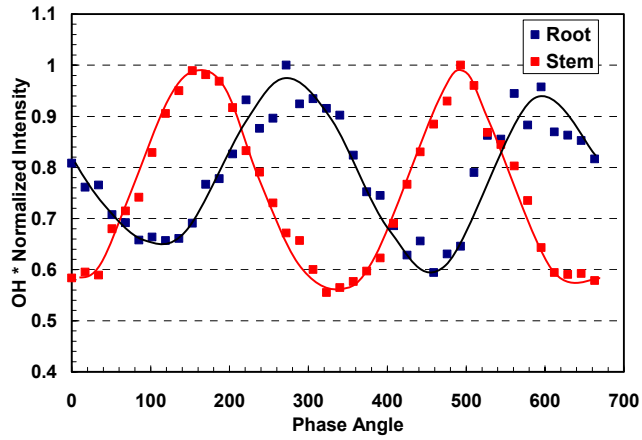
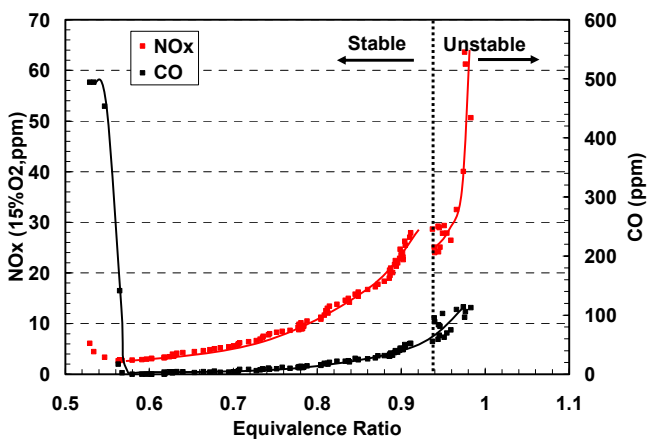
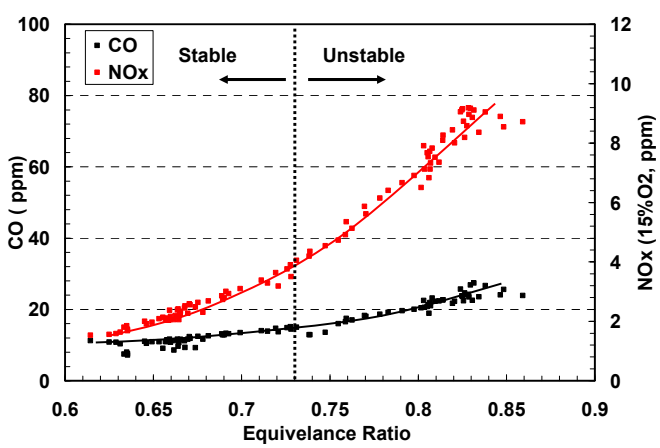


Figure 10: Periodic change of integrated OH* from root and stem parts versus phase angle: propane, L_{mt}=0.5, Case2.

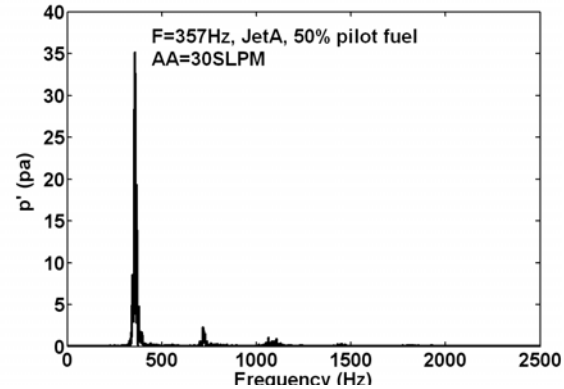


L_{mt}=0

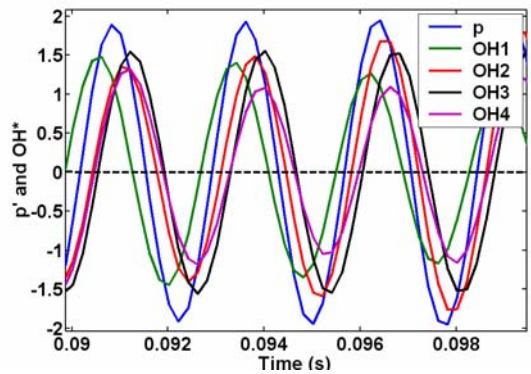


L_{mt}=0.5

Figure 11. Comparison of dependence of NO_x and CO emission on fuel equivalence ratio for cases 1 and 2: propane, L_{mt}=0 and L_{mt}=0.5.



(a)



(b)

Figure 9: Combustion instability at Case 3: Jet-A, L_{mt}=0. (a) FFT spectra of acoustic pressure p'; (b) Comparison of waveforms of acoustic pressure p' and OH* at four locations 1, 2, 3 and 4 (The waveforms shown here are filtered by band pass filters centered at 357Hz and the magnitudes of the signals are adjusted for purpose of comparison).

The effect of unstable combustion on NO_x and CO emissions for L_{mt}=0 and 0.5 is shown in Fig. 10. At L_{mt}=0, NO_x sharply jumped up when combustion transited from stable to unstable at Φ=0.94. But for L_{mt}=0.5, NO_x gradually increased with fuel equivalence ratio with no sudden transition. CO and NO_x of L_{mt} =0.5 was significantly reduced in the range of Φ = 0.65 – 0.85 compared with L_{mt} =0, highlighting the significant advantage of premixing in reducing NO_x emissions.

C. Case3&4: Liquid fuel with 50/50 fuel distribution at L_{mt}=0

Combustion instability was tested at 50/50 pilot and main fuel distribution case with Jet-A and Ethanol fuel, respectively. Figure 11 shows the

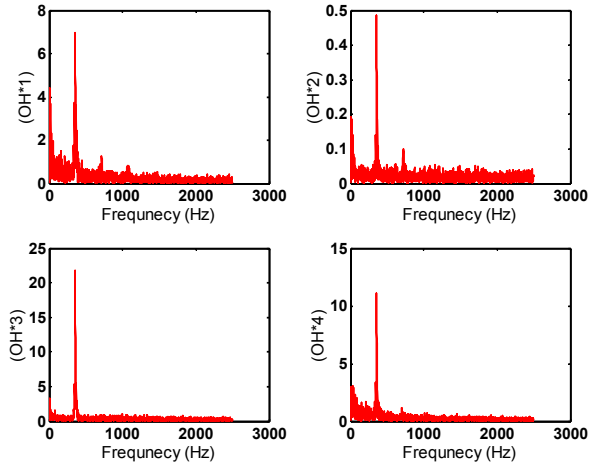
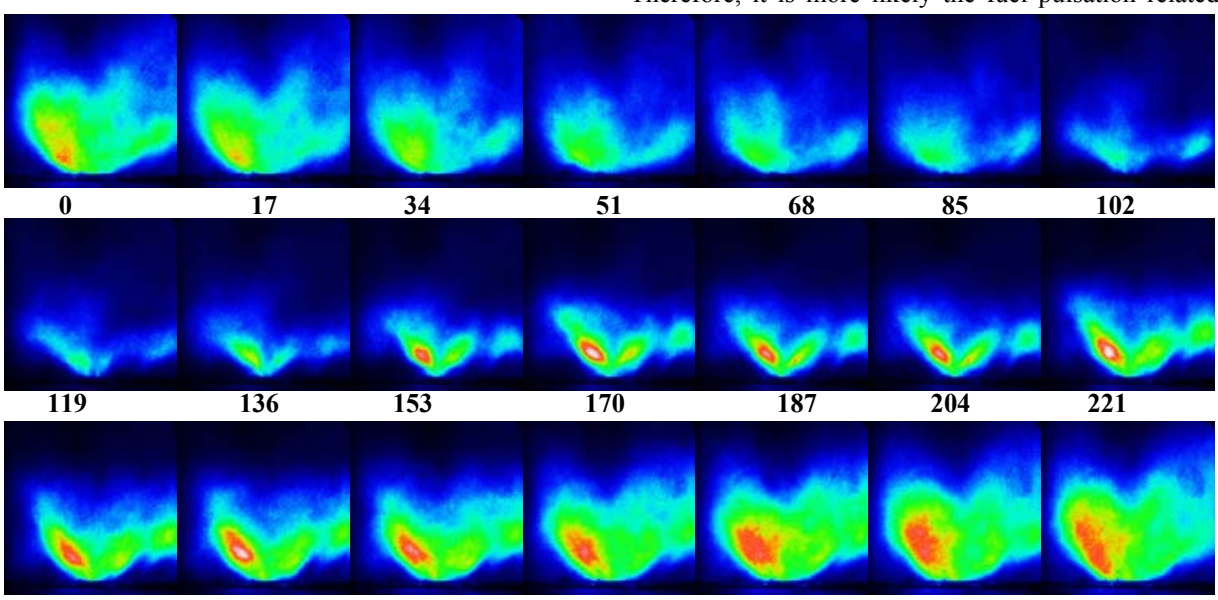


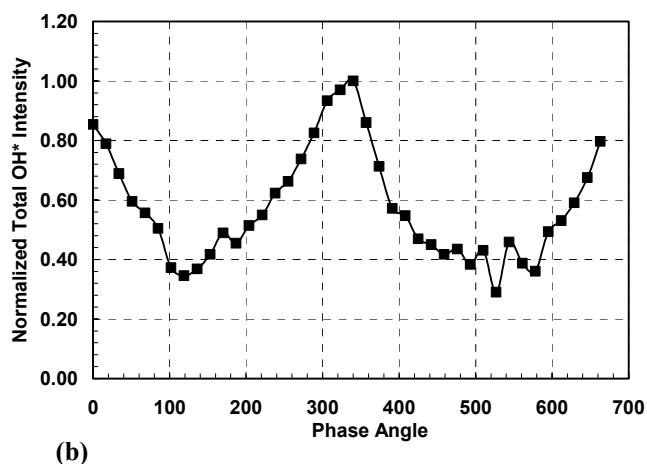
Figure 13. Spectra of OH* of unstable combustion at four locations: Case 3, Jet-A, L_{mt}=0.

spectra of p' and the waveforms of p' and OH* of four channels. The dominant frequency is 357Hz at Φ =0.6, with all signals approximately in phase. Spectra of OH* at four locations (Fig. 12) unanimously showed the same dominant frequency at 357 Hz.

Unlike gaseous fuel, the flame of spray combustion with Jet-A fuel is quite asymmetric. Figure 13a is a sequence of OH* chemiluminescence images within one cycle of instability for Case 3 (Jet-A fuel). The flame is anchored to the nozzle during the unstable combustion process, extending downstream along two branches with an intersection angle about 90°. Starting from the weakest flame at 119°, the reaction gets intense on the left side to form a strong reaction core. Intensity of this reaction core reduced when the flame further increases its area from 272° to 323º and then decreases into next cycle. During this instability cycle, no clear flame curvature was observed to claim vortex involved in this process. Therefore, it is more likely the fuel pulsation related to

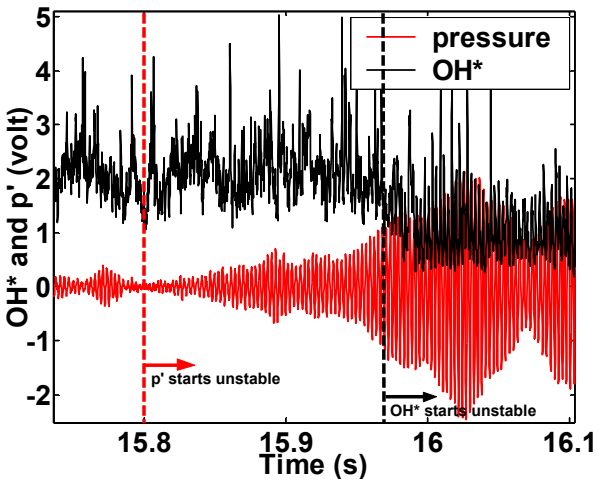


(a)

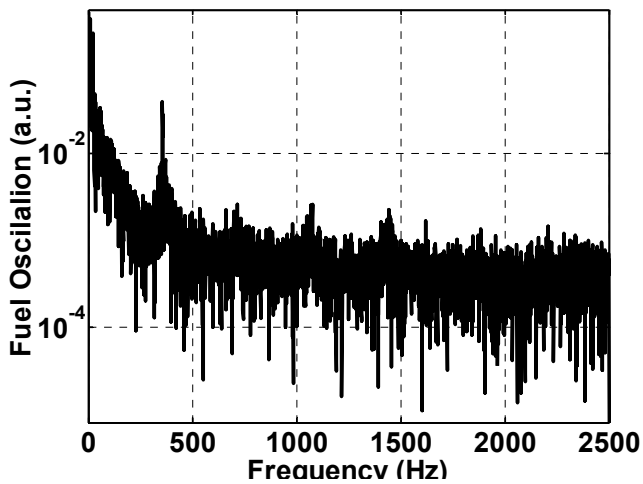


(b)

Figure 12: Oscillations of flame at unstable combustion for Case 3, Jet-A, L_{mt}=0: (a) Phase averaged OH* images at different phase angles within one cycle of unstable combustion. The unit of angle is degree. (b) Normalized total OH* intensity integrated from individual image versus phase angle.

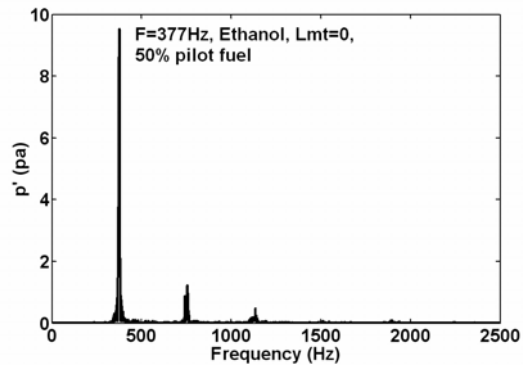


(a)

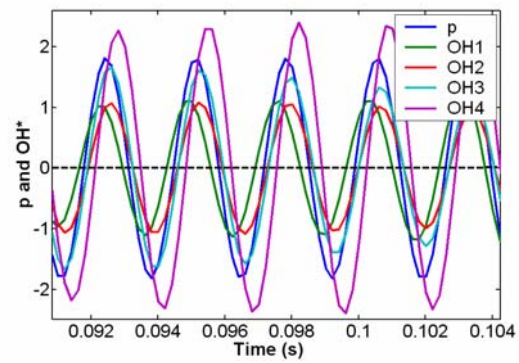


(b)

Figure 15. Signals of unstable combustion Case 3: Jet-A, L_{mt}=0. (a) Time trace of transition stage from stable to unstable combustion; (b) Spectra of oscillations in fuel supply line.



(a)



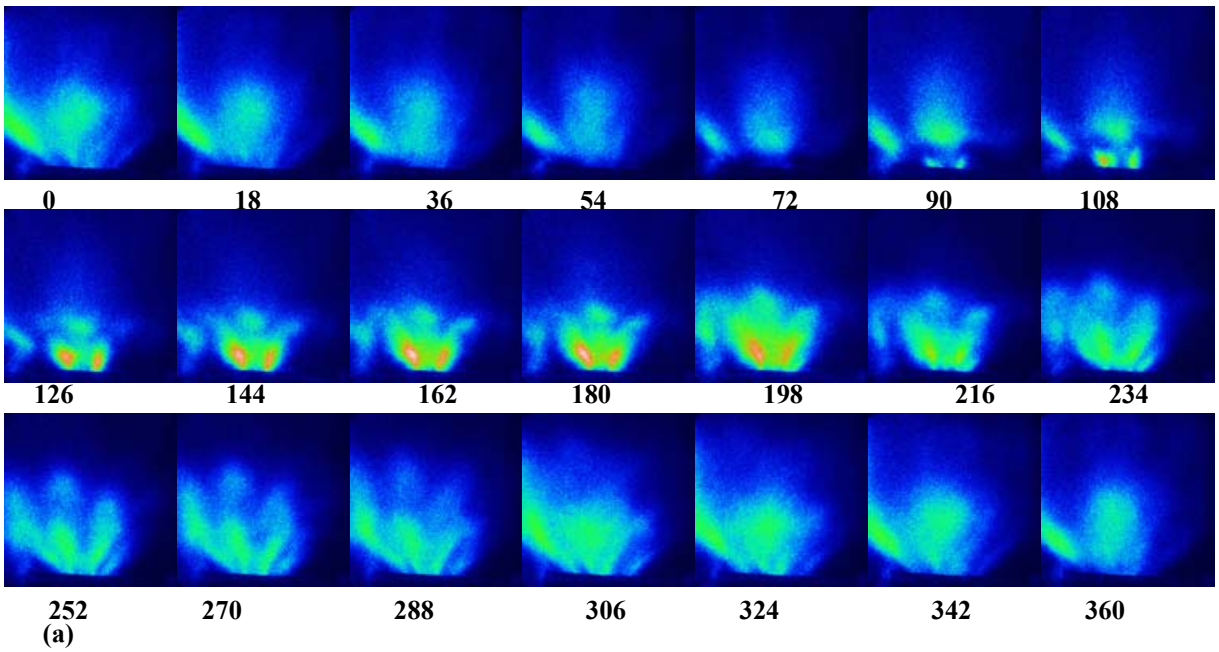
(b)

Figure 14: Combustion instability at Case 4: Ethanol, L_{mt}=0. (a) FFT spectra of acoustic pressure p'; (b) Comparison of waveforms of acoustic pressure p' and OH* at four locations 1, 2, 3 and 4 (The waveforms shown here are filtered by band pass filters centered at 357Hz and the magnitudes of the signals are adjusted for purpose of comparison).

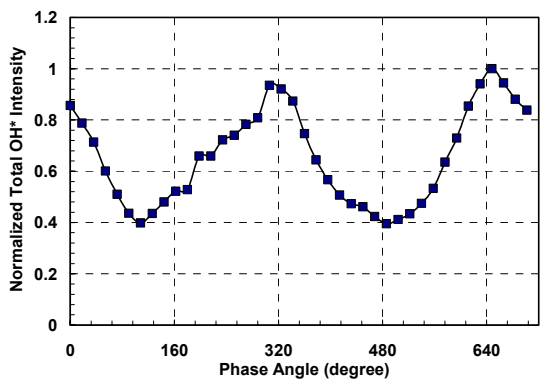
oscillation of fuel equivalence ratio causes periodic heat release and sustains combustion instability. Figure 13b shows the variation of total OH* intensity versus phase angle in two cycles.

The transition from stable to unstable combustion of case 3 is shown in Fig. 14a. The red and black lines marked the starting time of oscillations of pressure and flame, respectively. In this case, the initialization of oscillation of acoustic pressure led that of the flame about 180ms, much longer than case 1. It was also noticed that the flame extended to lean extinguish region (OH* signal approaching zero volt) when combustion became unstable, indicating weak flame or even flame out in certain time slots as observed by high speed flame imaging [7]. This is distinct from the transition in gaseous fuel case where the flame oscillated towards intense region. By monitoring the oscillations in liquid fuel supply line with a differential pressure transducer, we found a secondary peak at the instability frequency 357Hz (Fig. 14b). Combined with the time trace in Fig. 14a, we believe that the grownup of pressure oscillation gets into the fuel line and forces the fuel to pulse, resulting oscillation of fuel equivalence ratio that ultimately causes the large magnitude flame oscillation.

For combustion with ethanol (Case4), the spectra of p' and waveforms of p' and four OH* signals are shown in Fig.15a and b. For same operating condition as Case 3, the instability of Case 4 occurred at higher frequency 377Hz (Fig. 15a). Acoustic pressure and all OH* signals are in phase to indicate the axisymmetric longitudinal instability. Fig. 16a and b show a sequence of OH* images of Case 4 and the total OH* intensity versus phase angle, respectively. Different from Jet-A (Case3), two more flame branches are formed in addition to the central two. Also,



(a)



(b)

Figure 16: Oscillations of flame at unstable combustion for Case 4: ethanol, L_{mt}=0. (a) Phase averaged OH* images at different phase angles within one cycle of unstable combustion. The unit of angle is degree. (b) Normalized total OH* intensity integrated from individual image versus phase angle.

the central flame burns very close to the fuel nozzle, growing up and then moving toward downstream. It can be clearly seen that the combustion mainly occurs along the region where fuel sprays: it first concentrates on fuel stripes and then spreads out when fuel is vaporized and mixed with surrounding air. For gaseous fuel, no such distinct flame stripes can be discerned.

D. Dependence of p' and RMS OH* on equivalence ratio

The combustion instabilities studied here are all taking place when fuel is relatively rich. For instance, variation of oscillations of acoustic pressure and OH* radical emissions versus Φ is shown in Fig. 17 as the typical case. The combustion became unstable when the fuel flow rate increased to Φ=0.73 while air flow rate was kept at constant. Both p' and Root Mean Square (RMS) of OH* emission reached the maximum at Φ=0.8 (where the flame is most intense in terms of mean OH* emission also) and then dropped with fuel even richer.

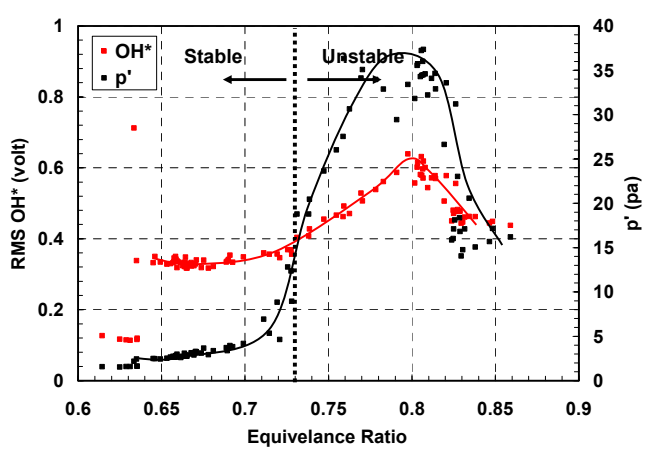


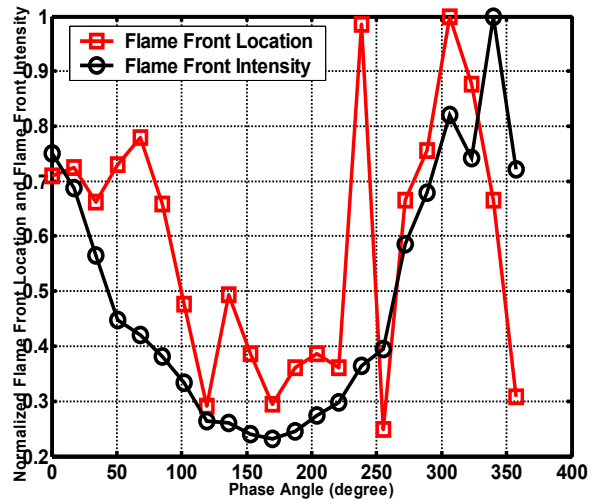
Figure 17: Dependence of oscillations of p' and OH* on the fuel equivalence ratio for Case 2: Propane L_{mt}=0.5.

E. Flame motions in unstable combustions

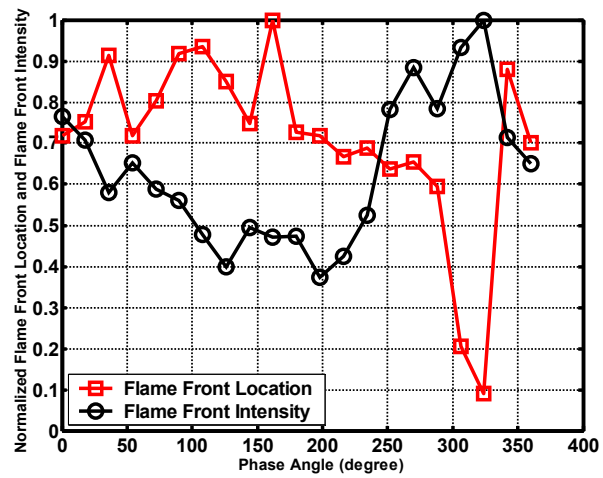
As illustrated in Fig. 5, the flame moved up and down periodically at the same phase with OH* intensity and acoustic pressure for gaseous fuel. It was argued that the formation of large vortical structure at the dump plane was responsible for this flame movement. By applying the same procedure, we also tracked the motion and intensity of flame front for liquid fuel cases. Figure 18a and b depicts the variation of flame front locations and flame front intensity versus phase angle for case 3 and 4, respectively. For unstable combustion with Jet-A fuel (Case 3), the flame front moves in phase with the variation of flame intensity on most time in one cycle. This harmony disappears for unstable combustion of ethanol fuel (Case 4) in which case the flame most time only slightly vibrates around the fairly fixed position (vertical coordinates 0.8 in Fig. 18b). Recalling the cases 3 and 4 were at same operating conditions with only difference in fuel types, we suspect that the fuel spray patterns of Jet-A and ethanol may play certain role in the coupling chain of unsteady heat release and pressure oscillation. The distinction of flame structure in Fig. 13a and Fig. 16a can be one evidence supporting this argument. Also, the quasi-steady flame front of case 4 indicates that pulsation of flame front may not be one necessary manifestation of unstable combustion. In cases with pulsation of flame front, namely, case 1 for gaseous fuel and case 3 for liquid fuel, the momentum of flame front motion may be associated with periodic large flow structures or pulsation in fuel supply line. In either case, the fact that oscillations of pressure led flame strongly suggests the response of flame front to acoustic pressure disturbance is one key factor in finally closing the coupling loop of unstable combustion. Candel [9] also argued pulsation of flame front was one of driving process of combustion instabilities among variety of mechanism involved dynamics of flow.

F. Growth rate of pressure oscillation p'

Figure 6 and Figure 14a illustrates the role of pressure as the trigger for transition from stable combustion to unstable combustion for gaseous fuel (Case1) and liquid fuel (Case3), respectively. To better understand the growth of p' in this transition region, we extract the magnitude of p' at this sampling range and sorting the valve from 0 to the unstable p' magnitude (amplitude of limit cycle). Figure 19 shows the comparison of growth rate of p' for the two types of fuel. Compared with spray combustion, the gaseous fuel combustion favors better mixing and promotes much faster growth rate of p': it takes 40 ms for p' reaching 15pa in gaseous



(a)



(b)

Figure 18: Variation of flame front location and flame front intensity on phase angle of unstable combustion cycle: (a) Case 3, L_{mt}=0, Jet-A; (b) Case 4, L_{mt}=0, Ethanol.

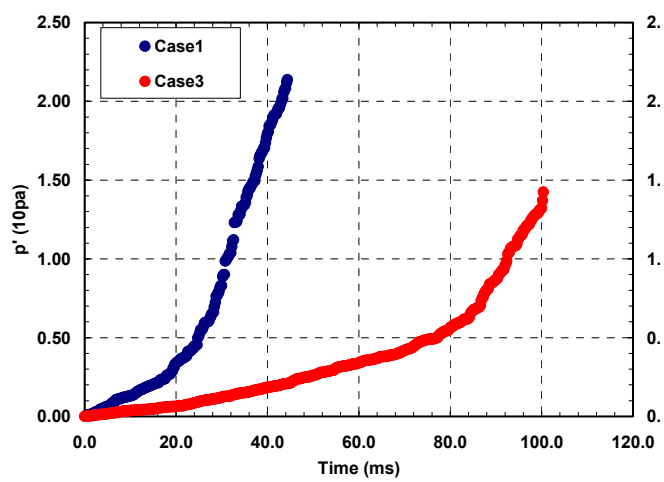


Figure 19: Comparison of growth rate of p' for gaseous fuel (Case 1) and liquid fuel (Case 3).

combustion (case1) whereas 100ms to get the same amplitude for spray combustion (Case3). This finding that well-mixed combustion has greater p' growth rate compared with diffusion spray combustion is very important for understanding combustion dynamics in well-mixed LDI combustion and premixed combustion.

IV. Conclusion

Manifestations of combustion instability for different geometries and types of fuel are revealed based on simultaneously measurement of acoustic pressure, local OH* radical emissions and phase-revolved OH* chemiluminescence imaging. All these combustion instabilities may be categorized as longitudinal mode because the dominant flame oscillation is in the axial direction of the combustor and all optical signals from circumferentially installed fibers are approximately in phase. The acoustic pressure and unsteady heat release, which is reflected on local and global radical emissions, are strongly coupled when unstable combustion takes place for all the four cases. The dynamics of flame, however, are shown to in different forms depending on the fuel types and existence of premixing section. Pulsation of flame front is associated with unstable combustion that is involved in periodic evolution of large flow structures, which is evidenced by the curvature of flame with the gaseous fuel case, or pulsation in fuel supply that occurred for Jet-A case. Also, the pulsation of flame front is coincident with the observation that oscillations of pressure lead the flame.

Mixing tube promotes combustion instability to occur at leaner equivalence ratio. The specific flame motion in mixing tube case may indicate the mixing section a host for unsteady combustion. Although good mixing helps in reducing NO_x and CO emissions, the mach faster unstable combustion growth rate offsets this advantage and suggests this issue need to be carefully addressed in applying premixing combustion for NO_x reduction.

References

[1]Mongia, H. C., Held, T. J., Hsiao, G. C., and Pandalai, R. P., "Challenges and Progress in Controlling Dynamics in Gas Turbine Combustors", *Journal of Propulsion and Power*, Vol. 19, No. 5, 2003, pp.822-829.

[2]Ducruix, S., Schuller, T., Durox, D., and Candel, S., "Combustion Dynamics and Instabilities: Elementary Coupling and Driving Mechanisms", *Journal of Propulsion and Power*, Vol. 19, No. 5, 2003, pp.722-734.

[3]Poinsot, T. J., Trouve, A. C., Veynante, D. P., Candel, S. M. and Esposito, E. J., "Vortex-driven Acoustically Coupled Combustion Instabilities", *Journal of Fluids Mechanics*, Vol. 177, 1987, pp.265-292.

[4]Paschereit, O., Gutmark, E., Weisenstein, W., "Excitation of Thermoacoustic Instabilities by Interaction of Acoustics and Unstable Swirling Flow," *AIAA Journal*, Vol. 38, No. 3. 2000.

[5]Broda. J. C., Seo, S., Santoro, R. J., Shirhattikar, G., and Yang, V., "An Experimental Study of Combustion Dynamics of a Premixed Swirl Injector," *Proceeding of 27th International Symposium on Combustion/The Combustion Institute*, 1998, pp.1849-1856.

[6]Giezendanner, R., Keck, O., Weigand, P., Meier, W., Meirt, U., Stricker, W., and Aigner, M., "Periodic Combustion Instabilities in a Swirl Burner Studied by Phase-locked Planar Laser-induced Fluorescence," *Combustion Science and Technology*, Vol. 175, 2003, pp.721-741.

[7]Ng, W. B., Clough, E., Syed, K. J., and Zhang, Y., "The Combined Investigation of the Flame Dynamics of an Industrial Gas Turbine Combustor Using High-speed Imaging and an Optically Integrated Data Collection Method," *Measurement Science and Technology*, Vol. 15, 2004, pp. 2303–2309.

[8]Li, G., and Gutmark, E. J., "Experimental Study of Large Coherent Structure in a Swirl Dump Combustor", *42nd AIAA Science Meeting and Exhibit*, AIAA-2004-0133, Reno, NV, 2004.

[9]Candel, S. M., "Combustion Instability Coupled by Pressure Waves and Their Active Control," *24th Symposium (International) on Combustion/The Combustion Institute*, 1992, pp. 1277-1296.

[10]Li, G., Gutmark, E, J., "Geometry effects on the flow field and the spectral characteristics of a Triple Annular Swirler", *Proceeding of 2003 International Gas Turbine Institute*, GT-2003-38799, 2003.